Part 1: Cache Memory & Organization

1. ________ is the time interval between the request for information and the access to the first bit of that information. a) Cycle Time b) Latency (Access Time) c) Bandwidth d) Transfer Rate

• **Answer:** b) Latency

2. Which cache mapping technique is the most expensive to implement because it requires comparators for all blocks? a) Direct Mapping b) Set Associative c) Fully Associative d) Block Mapping

• **Answer:** c) Fully Associative,

3. In ________ mapping, each block of main memory maps to only one specific line in the cache. a) Direct b) Associative c) Set Associative d) Indirect

• **Answer:** a) Direct,

4. If the Hit Time is 5 ns, Miss Rate is 10%, and Miss Penalty is 100 ns, what is the Average Memory Access Time (AMAT)? *(Formula:)* a) 10 ns b) 15 ns c) 105 ns d) 20 ns

• **Answer:** b) 15 ns

5. Which replacement policy removes the block that has been in the cache for the longest time regardless of use? a) LRU b) LFU c) FIFO d) Random

• **Answer:** c) FIFO

6. Consider a Direct Mapped cache of size 512 KB with a block size of 1 KB. How many bits are required for the ? *(Calculation: bytes)* a) 8 b) 9 c) 10 d) 12

• **Answer:** c) 10

7. Using the same data (Cache 512 KB, Block 1 KB), how many bits are required for the ? *(Calculation: Number of lines =)* a) 8 b) 9 c) 10 d) 12

• **Answer:** b) 9

8. ________ is the time elapsed from the start of a read operation to the start of a subsequent read. a) Cycle Time b) Latency c) Access Time d) Seek Time

• **Answer:** a) Cycle Time

9. Which cache mapping function has the highest probability of "Collision" or "Conflict"? a) Fully Associative b) Set Associative c) Direct Mapping d) Indirect Mapping

• **Answer:** c) Direct Mapping

10. L1 cache is generally ________ than L3 cache. a) Larger and Slower b) Smaller and Faster c) Larger and Faster d) Smaller and Slower

• **Answer:** b) Smaller and Faster

Part 2: Assembly Language & Registers

11. In assembly language, the _________ register is specifically used for looping. a) AX b) BX c) CX d) DX

• **Answer:** c) CX

12. Which of the following instructions typically takes operands? a) MOV b) SHRD c) INC d) RET

• **Answer:** b) SHRD,

13. In assembly language, _________ explain the program's purpose and are ignored by the assembler. a) Directives b) Identifiers c) Comments d) Mnemonics

• **Answer:** c) Comments,

14. Which of the following is a typical assembly language instruction (it is a directive)? a) MOV b) ADD c) PAGE d) SHRD

• **Answer:** c) PAGE

15. Which register keeps track of the address of the next instruction to be executed? a) IR b) MAR c) Program Counter (PC/IP) d) Accumulator

• **Answer:** c) Program Counter (PC/IP),

16. All of the following are reserved words in assembly language EXCEPT: a) SIZE b) MODEL c) FAR d) FLDD

• **Answer:** d) FLDD (Note: SIZE, FAR, MODEL are reserved),

17. Which of the following is a label name? a) 1start b) _Start c) MOV d) @123

• **Answer:** b) _Start (Labels cannot start with numbers like "1start"),

18. The instruction RET (Return) has _________ operands. a) 0 b) 1 c) 2 d) 3

• **Answer:** a) 0

19. Which register is used to hold the instruction currently being executed (decoded)? a) PC b) IR c) MAR d) MBR

• **Answer:** b) IR

20. In Signed Integers, if the highest bit (MSB) is 1, it indicates that the number is _________. a) Positive b) Negative c) Zero d) Invalid

• **Answer:** b) Negative,

Part 3: System Bus & Interrupts

21. _________ interrupts occur due to illegal operations (like division by zero) or erroneous use of data. a) External Interrupts b) Traps (Internal Interrupts) c) I/O Interrupts d) Hardware Interrupts

• **Answer:** b) Traps (Internal Interrupts),

22. Which bus setup is faster and well-suited for applications where messages are generated at irregular intervals (e.g., keystrokes)? a) Synchronous b) Asynchronous c) Dedicated d) Time Multiplexed

• **Answer:** b) Asynchronous,

23. In a _________ bus, the timing of data transfers is determined by a global clock signal. a) Asynchronous b) Synchronous c) Handshaking d) Semi-Asynchronous

• **Answer:** b) Synchronous

24. The _________ Bus is bi-directional to allow flow in both directions. a) Address b) Control c) Data d) Power

• **Answer:** c) Data

25. Which arbitration method allows the devices themselves to determine who has the highest priority using a common line? a) Centralized b) Distributed c) Daisy Chain d) Polling

• **Answer:** c) Daisy Chain

Part 4: General Concepts & Conversions

26. The binary equivalent of the decimal number (37) is: *(Calculation: 32 + 4 + 1 = 100101)* a) 11101111 b) 00100101 c) 00111111 d) 00011111 (Note: Check specific conversion in exam key if available, standard conversion is 100101. Based on source for (37) octal or similar, ensure base is checked. *Correction: Source asks for (233)10. Let's use a standard conversion.*) **Replaced Question based on Source: The binary equivalent of $(233)_{10}$ is:** a) 11101001 b) 00100101 c) 10101010 d) 11001100

• **Answer:** a) 11101001

27. Architecture describes _________ the computer does, while Organization describes _________ it does it. a) How, What b) What, How c) When, Where d) Why, When

• **Answer:** b) What, How

28. Which of the following is NOT a characteristic of Main Memory (RAM)? a) It is Volatile b) It requires refreshing (if DRAM) c) It is non-volatile d) It is slower than Cache

• **Answer:** c) It is non-volatile

29. The hexadecimal equivalent of $(125)_{10}$ is: *(Calculation: rem)* a) 7D b) D7 c) 77 d) DD

• **Answer:** a) 7D

30. ________ describes the number of instructions per second the processor can execute. a) Bandwidth b) Latency c) Throughput d) Capacity

• **Answer:** c) Throughput (or Processor Bandwidth in some contexts)